Canine Cognitive Dysfunction

Pathophysiology, Diagnosis, and Treatment

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KEYWORDS

• Canine • Cognitive • Dysfunction • Dementia • β-Amyloid • Vascular • Alzheimer disease

KEY POINTS

- Canine cognitive dysfunction (CCD) is the canine analog of human Alzheimer disease (AD).
- The pathophysiology of CCD/AD is multifaceted and involves brain vascular damage, deposition of toxic β-amyloid protein in the brain, oxidative brain injury, neuronal mitochondrial dysfunction, excitotoxic neuronal damage, and inflammation.
- CCD is common in aged (>8 years of age) dogs, affecting between 14% and 35% of the pet dog population.
- Apparent confusion, anxiety, disturbance of the sleep/wake cycle, and decreased interaction with owners are all common clinical signs of CCD.
- Although there is no cure for CCD, several proven effective therapeutic approaches are available for improving cognitive ability and maintaining a good quality of life; instituting such therapies early in the disease course is likely to have the greatest positive clinical effect.

INTRODUCTION

Canine cognitive dysfunction (CCD) is the canine analog of human Alzheimer disease (AD). CCD is common in older dogs, particularly those more than 8 years old. Slowly progressive signs of altered mentation and dementia characterize the disorder. The

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pathophysiology of CCD/AD involves brain vascular disease and accumulation of β-amyloid (Aβ) protein; these two processes are intertwined, each promoting the progression of the other. Aβ is a neurotoxic protein that accumulates in the brains of dogs with CCD and people with AD and forms plaques within the brain parenchyma. Accumulation of Aβ also contributes to cerebrovascular disease, referred to as cerebrovascular amyloid angiopathy (CAA).[1–6] Other pathologic processes that contribute to cognitive impairment in CCD/AD include oxidative brain damage, neuronal mitochondrial dysfunction, glutamate-mediated excitotoxic neuronal damage, impaired neuronal glucose metabolism, and proinflammatory processes that stem from abnormal microglial and astrocyte function.[1–11] The 3 most common misconceptions regarding CCD are that mild cognitive impairment reflects normal canine aging, CCD is not common, and that there are no effective treatments for the condition. Despite pathologic similarities between CCD and AD, dogs with CCD typically do not show the severe cognitive impairment that is experienced in patients with AD.[1,12] This article reviews the pathophysiology of CCD, as well as characteristic features and available treatment options. Most of the information in this article is based on currently available literature regarding CCD and AD, but the authors' collective clinical experience with CCD is also provided.

CLINICALLY RELEVANT PATHOPHYSIOLOGY AND GENERAL CONSIDERATIONS

CCD is an age-related disorder similar to AD in people that occurs in elderly dogs. Age-related cognitive dysfunction has been extensively studied in the dog, and this species seems to be the best naturally occurring animal model available for human AD. Estimates of the prevalence of CCD generally vary between 14% and 35% of the pet dog population. Many of the reported estimates of prevalence likely underestimate how common the disorder is. As with people with AD, the prevalence of CCD increases dramatically with age. In one study, the prevalence of CCD in dogs 11 to 12 years old was 28%, and 68% in dogs 15 to 16 years of age.[1,3,12–14] In a 2-year prospective longitudinal study of dogs more than 8 years of age, 33% of dogs with normal cognitive status progressed to mild cognitive impairment, and 22% of dogs with mild cognitive impairment progressed to CCD.[15]

There are pathologic similarities between the brains of people with AD and dogs with CCD (**Box 1**).

Cerebral vascular changes, meningeal thickening, gliosis, and ventricular dilatation occur in brains of both patients with AD and with CCD. More specifically, progressive

Box 1
Pathologic brain abnormalities identified in both people with Alzheimer disease and dogs with canine cognitive dysfunction

Cerebrovascular disease
Aβ accumulation
Oxidative brain damage
Neuronal mitochondrial dysfunction
Glutamate-mediated excitotoxic neuronal damage
Impaired neuronal glucose metabolism
Microglial dysfunction
Astrocyte dysfunction

accumulation of the neurotoxic protein Aβ in the brain (around neurons and blood vessels) is a consistent feature in both AD and CCD. These accumulations coalesce to form plaques (neuritic plaques) and are most prominent in the frontal cerebral cortex and hippocampus in both human and canine disorders. The most common form of insoluble Aβ found in brain tissue is a 42-amino-acid form, called $A\beta_{42}$. In addition to this insoluble form of the protein, smaller soluble oligomeric forms of Aβ have been identified; these are highly toxic and interfere with synaptic function.[1–4,6,16–19] In both human and canine disorders, the degree of Aβ accumulation (Aβ load) has been correlated with the extent of cognitive impairment.[20] However, this is not uniformly accepted, and there is recent evidence that the relationship between Aβ load and cognitive impairment is tenuous at best. In one histopathologic study of dogs with CCD, cognitive impairment was associated with increased numbers of astrocytes and microglial cells, as well as increased ubiquitin protein levels, rather than with levels of Aβ.[21] In another histopathologic study, increasing Aβ load was a function of age rather than level of cognitive impairment; in this study, the type of Aβ plaques was similar to that seen in early stages of human AD.[11] In addition to the accumulation of neurotoxic Aβ protein in the aged canine brain, intraneuronal accumulation of a hyperphosphorylated microtubular-associated protein (tau protein) has also been shown. Tau protein is the precursor to neurofibrillary tangles (NFTs), another prominent histopathologic feature of human AD. The absence of mature NFTs in the brains of dogs with age-related cognitive dysfunction has been argued as evidence against the canine disorder being analogous to human AD. However, there are several potential explanations to account for the absence of NFTs in dogs with CCD. It is possible that dogs do not live long enough for the tau proteins to develop into NFTs as they do in people. Although the amino acid sequence of Aβ protein is identical in people and dogs, this is not the case for tau protein. Because the amino acid sequence of dog tau protein differs from that in people, this may affect the ability of tau protein to form NFTs in the former species. Other structural abnormalities (**Box 2**) found in the aging canine brain that are similar to those in humans include cerebral atrophy, ventricular enlargement, blood vessel wall fibrosis, amyloid deposition (meningeal and parenchymal), microhemorrhages (and occasionally macrohemorrhages) and infarcts, axonal degeneration with myelin loss, astroglial hypertrophy and hyperplasia, and intraneuronal accumulation of several substances (lipofuscin, polyglucosan bodies, and ubiquitin).[1,2,4–6,11,16–18,22]

The pathophysiology of CCD and AD is multifactorial and complex. In addition to Aβ deposition and vascular disease, numerous cellular and biochemical aberrations contribute to progressive cognitive decline. There is evidence in both diseases that

Box 2 **Structural brain abnormalities identified grossly and on MRI in both people with Alzheimer disease and dogs with canine cognitive dysfunction**
Cerebrovascular disease
Infarcts
Microhemorrhages and occasionally macrohemorrhages
Cerebral atrophy
Meningeal thickening
Ventricular dilatation
Gliosis

increased oxygen free radical–mediated cellular damage, decreased endogenous antioxidant defenses, decreased mitochondrial function, inflammation (from various processes), DNA damage, vascular compromise, and neurotransmitter imbalance are all interrelated processes that are involved in progressive cognitive impairment. Neurochemical changes occur in the aging brain that also contribute to progressive cognitive impairment. Age-associated decline in the brain neurotransmitter levels of acetylcholine, dopamine, norepinephrine, and gamma-aminobutyric acid (GABA) are documented in CCD and AD. Of these abnormalities, cholinergic dysfunction seems to have the highest and most consistent correlation with age-related cognitive impairment. Other neurochemical abnormalities identified in brains of patients with AD and with CCD include increased acetylcholinesterase levels (associated with cholinergic decline), increased monoamine oxidase B levels (catalyzes the breakdown of dopamine, with subsequent formation of free radicals), and increased cerebrospinal fluid (CSF) levels of lactate, pyruvate, and potassium.

Mitochondrial dysfunction is a central theme in the development and progression of AD/CCD. Mitochondria produce ATP via the electron transport chain and are responsible for providing most of the cellular energy requirements. In AD/CCD, mitochondria undergo both morphologic and functional changes (including mitochondrial gene expression), some of which are attributed to Aβ deposition. In general, mitochondrial function declines in AD/CCD. In addition to their role in cellular energy production in brain cells, the mitochondria have a proteostatic ability to decrease the toxic effect of Aβ on cellular function that diminishes as mitochondrial function declines. In addition, and related to, decreased mitochondrial ability to generate cellular energy, the CCD/AD brain has a diminished capacity to metabolize glucose. Glucose is the main energy substrate for neurons in the brain, and neurons affected by AD/CCD are impaired in both uptake and use of glucose.[1–11,23,24]

An imbalance in the levels of neuroexcitatory versus neuroinhibitory neurotransmission in the brain has been documented in animal models and patients with AD. Neuronal hyperexcitability is thought to contribute to cognitive decline in patients with AD. Excessive stimulation of glutamate receptors in AD/CCD contributes to neuronal excitotoxicity and death. In addition, there is evidence that impaired function of inhibitory interneurons is an important facet of abnormal neural networks in the AD brain. These abnormalities of synaptic plasticity have been correlated with Aβ deposition and mitochondrial dysfunction.[4,7,8]

As previously mentioned, there are several misconceptions regarding CCD that are often held by both dog owners and members of the veterinary community. These misconceptions likely contribute to misdiagnosis and delayed diagnosis of CCD, as well as undertreatment of the disorder. Pet owners often regard development of clinical signs of cognitive dysfunction, especially subtle early signs, as part of the normal aging process. CCD is a common disorder, and it often progresses from mild to more severe cognitive impairment. As cognitive ability progressively declines in patients with CCD, it is common for owners to assume that the outlook is hopeless and that nothing can be done. Although there are striking similarities between CCD and AD in terms of pathophysiologic characteristics and mechanisms, there are some important differences regarding disease severity, rate and extent of progression, and societal roles of the affected individual. Dogs with CCD tend to have less severe levels of Aβ-associated brain disorder than are observed in patients with AD, and are less likely to develop such severe impairment as occurs in people with AD.[1,11,14,15] Patients with CCD typically respond well to medical intervention, especially if instituted early in the disease process, and the interventions typically applied have little to no adverse effect. There is also evidence in both AD and CCD that preventive measures (eg, diet

changes, environmental enrichment) can both delay the onset and slow the progression of cognitive decline.[1,3,24,25] Overall, successful management of patients with CCD entails prolongation of a good quality of life of the patient as an interactive family pet. Successful management of patients with AD is much more complex with respect to the patients retaining the same role in society as before the development of the disorder. Although it is useful to compare disease mechanistic processes between AD and CCD, veterinarians should also view CCD as a distinct disorder in its own right when considering treatment options and prognosis.

In consideration of the potential benefits of early intervention in CCD management, note that clinically inapparent cognitive decline has been documented to occur in some dogs before 8 years of age. Using specific behavioral testing tools, early cognitive impairment has been documented in beagles as young as 6 years old.[1,3,6,11–15] The same type of testing used in another study showed that young to middle-aged (1.5–7 years old) beagle dogs with increased CSF Aβ concentrations showed learning deficits, compared with dogs with low CSF Aβ concentrations.[25] Histopathologically observable brain Aβ deposits do not develop in dogs until age 8 or 9 years.[16,25] This information suggests that simple preventive measures against CCD (eg, dietary supplements, as discussed later) may be generally advisable in pet dogs as they near middle age.

SIGNALMENT, HISTORY, AND CLINICAL SIGNS

The typical scenario of CCD is an elderly dog (often >8 years of age) with slowly progressive cognitive decline, usually over a period of at least several months. In addition to advancing age, other reported risk factors for developing CCD include sex (females at highest risk, spayed/neutered at higher risk than intact dogs) and body size (smaller dogs at higher risk than larger dogs); however, other reports have not found associations between sex or body size and the likelihood of developing CCD.[1–4,6,11–14,24,26,27] One study found a significant association between idiopathic epilepsy and early-age onset of CCD in dogs; this is similar to people, in which there is an established relationship between temporal lobe epilepsy and early-onset AD.[28,29] The 4 main historical (ie, derived from the owner) clinical features of cognitive impairment are apparent confusion, anxiety, disturbance of the sleep/wake cycle (sleeping during the day, restless at night), and decreased interaction of the pet with the owners **Box 3**.[26]

In addition to these main features of CCD, historical complaints regarding the disorder are numerous and often nonspecific (**Box 4**). They include inattentiveness, inactivity, aimless wandering (often pacing at night), demented behavior, urinary and/or fecal incontinence (posturing normally, but voiding in inappropriate locations), difficulty navigating stairs, attempting to pass through narrow spaces (like the hinge region of a door), inability to locate dropped food, becoming lost in previously familiar environments, failure to recognize previously familiar people or animals, decreased interaction with family members, apparent hearing loss, and excessive vocalization

Box 3 **Four main owner observations given in history**
Apparent confusion
Anxiety
Disturbance of sleep/wake cycle
Decreased interaction of the pet with owners

Box 4 Other owner-reported behaviors commonly displayed by dogs with canine cognitive dysfunction
Inattentiveness
Inactivity
Compulsive wandering and pacing, especially at night
Demented behavior
Urinary and/or fecal incontinence (posture normally, but void inappropriately)
Difficulty navigating stairs
Attempting to pass through inappropriately narrow spaces
Unable to locate dropped food
Getting lost in familiar environments
Unable to recognize familiar people or animals
Decreased interaction with family
Apparent hearing loss
Excessive vocalization especially at night
Acting senile

(often at night). Owners of pets with CCD often describe their pets as acting senile.[1–4,6,12–15,26]

Dogs with CCD typically show evidence of forebrain dysfunction on clinical examination (**Box 5**). These patients have abnormal mentation and often respond inappropriately to their environment (dementia). Many dogs with CCD circle constantly in the examination room and either do not respond or respond inappropriately to visual and auditory stimuli. In the authors' experience, these dogs typically appear very anxious and tend to resist restraint. Also based on the authors' experience, dogs with suspected CCD occasionally show either transient central vestibular dysfunction or seizure activity of recent onset. Although not well described as a clinical feature of CCD, vestibulocerebellar dysfunction and seizures are reported as a potential consequence of AD in people.[4,30,31]

Box 5 Clinical signs associated with the forebrain that may be displayed by dogs with canine cognitive dysfunction
Anxiety
Abnormal mentation
Compulsive circling
Absent or inappropriate response to visual/auditory stimuli
Excessively resistant to even mild restraint
Transient vestibular episodes
Recent onset of possible seizure activity

DIAGNOSIS

In most cases, a CCD diagnosis is based on signalment, history, and clinical features that are all consistent with the diagnosis. Several useful and accurate behavioral tests have been developed that can objectively measure canine cognitive ability and deficits thereof in a laboratory setting. These tests include the delayed nonmatching to position (DNMP) memory task, and the attention task. The DNMP task evaluates short-term visuospatial memory, and the attention task tests selective attention.[1,3,4] Such tests have been, and remain, essential tools for CCD research but are not practical in a busy clinical setting. Other, more rapidly performed, cognitive tests (a food-searching [FS] task and problem-solving task) have been evaluated for clinical use, with the FS task showing the most promise.[32] Pivotal to a correct diagnosis of CCD is an accurate and comprehensive history from the pet owner. Clinicians must have a high index of suspicion for CCD in older dogs. In addition, specific questions regarding the patient's behavior need to be posed to the dog owner with suspected CCD. Owners often do not volunteer potentially useful historical behavioral observations, either because they do not associate them with the overall clinical picture or because they attribute them to signs of normal dog aging.

The only imaging modality of practical use for the diagnosis of CCD is MRI. Owners of dogs with CCD often elect not to pursue MRI, because of several factors. These factors include concerns over general anesthesia in a geriatric pet, costs associated with MRI, and the low likelihood that an MRI diagnosis will contribute meaningfully to the treatment plan. Because of this understandable reticence of owners to pursue MRI in dogs suspected of having CCD, the literature on MRI features of CCD is sparse compared with that available for AD. Brain imaging of patients with AD can be normal or may reveal brain atrophy, ventricular enlargement, and lesions in the medial temporal lobes of the cerebral cortex. Age-related changes appreciated on MRI of the brain in patients with CCD are primarily reflective of brain atrophy and include ventricular enlargement as well as widened and well-demarcated cerebral sulci. Although these are consistent findings associated with the aging brain, they can also be found in some older patients without evidence of CCD.[4] Periventricular white matter hyperintensities, also referred to as leukoaraiosis, are thought to be caused by vascular abnormalities of the arterioles in this region of the brain; in people, they have been associated with AD and other vascular brain disorders. Similar periventricular hyperintense regions have also been described in elderly dogs (**Fig. 1**).

In one report of 14 elderly dogs with suspected leukoaraiosis on MRI, a progressive decrease in interthalamic adhesion thickness was shown for 3 dogs for which serial

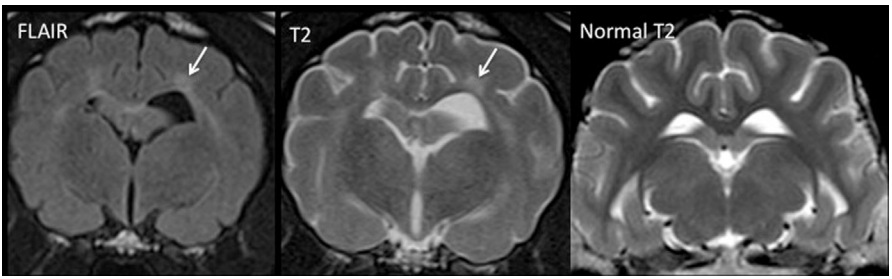

Fig. 1. Transverse fluid-attenuated inversion recovery (FLAIR) and T2-weighted (T2w) MRI of normal and affected brain showing bilateral hyperintense periventricular white matter lesions, referred to as leukoaraiosis (*white arrows*).

imaging was available.[33,34] The thickness of the interthalamic adhesion as measured on transaxial T1-weighted and T2-weighted MRI was significantly smaller in dogs with CCD compared with dogs without CCD; an interthalamic adhesion thickness of 5 mm or less was found to be consistent with a diagnosis of CCD (**Fig. 2**).[35]

A more recent study confirmed the diagnostic utility of measuring the thickness of the interthalamic adhesion in CCD; this study also found that the ratio of the interthalamic adhesion thickness to the brain height and the ratio of this number to the ratio of the lateral ventricle height to brain height were accurate predictors of CCD.[36]

Spontaneous intraparenchymal brain hemorrhage has been documented in both AD and CCD. Although MRI evidence of microhemorrhages is common in patients with AD, the prevalence of this finding in dogs with CCD has not been specifically investigated. The use of T2*-weighted MRI sequences has been shown to be useful in identifying hemorrhagic brain lesions in both people and dogs. In people with AD, MRI evidence of microhemorrhages on T2*-weighted imaging is common and is attributed to CAA. Occasionally, macrohemorrhages are also evident on T2*-weighted images in these patients.[5,37–43] In one large retrospective investigation of brain microhemorrhages in dogs that were imaged with T2*-weighted MRI sequences, older dogs of smaller breeds were significantly more represented. This study also found that these dogs were significantly more likely to present for vestibular dysfunction.[43] In the authors' experience, microhemorrhages are typically evident on T2*-weighted images of elderly dogs' brains that are imaged for recent onset of vestibular dysfunction, suspected seizure activity, and/or progressive behavioral abnormalities. We have also documented macrohemorrhages in several elderly dogs, with no obvious underlying metabolic cause for spontaneous intracranial bleeding (**Fig. 3**).

Laboratory findings are typically normal in dogs with cognitive dysfunction syndrome (CDS), unless there is an age-related concurrent disorder (eg, increased blood urea nitrogen/creatinine levels from chronic renal disease). In one study, plasma concentrations of $A\beta_{42}$ were significantly increased in dogs with CCD, compared with older dogs with either mild cognitive impairment or no cognitive impairment.[15] Because patients with suspected CCD are generally older, and the clinical features of CCD are numerous and often nonspecific, it is important to rule out geriatric

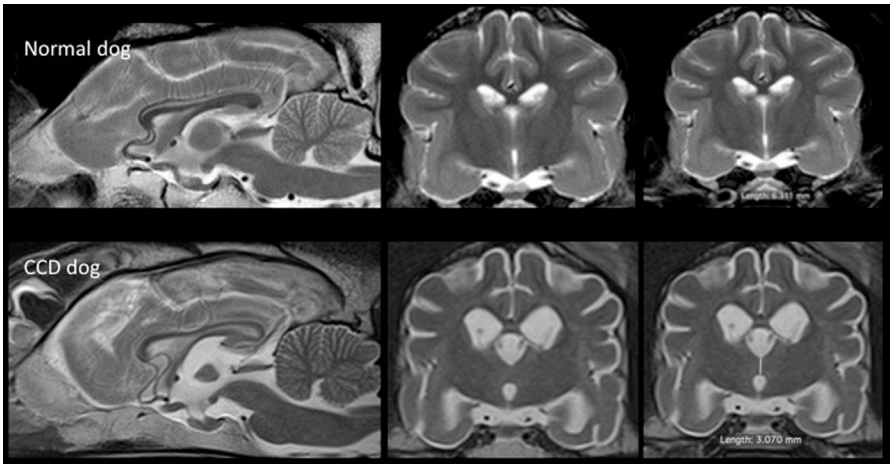

Fig. 2. Sagittal and transverse T2w MRI of normal and affected brain showing the reduced size and abnormal shape of the interthalamic adhesion.

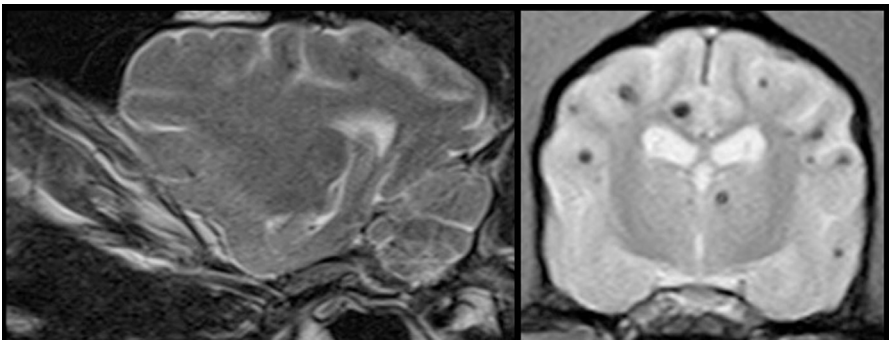

Fig. 3. Sagittal T2w and transverse T2*(fast field echo) MRI of affected brain showing multiple hypointense nonsymmetric bilateral punctate lesions affecting the forebrain.

disorders that may mimic signs of cognitive impairment. In particular, a differential diagnosis of a forebrain tumor is typically a top consideration in an elderly dog with behavioral changes. Organ dysfunction (eg, liver, kidneys) and pain may also lead to clinical signs that may be confused with cognitive impairment. In addition to confounding disease states that may mimic clinical features of cognitive impairment, it is important to know what drugs a patient is being administered, because some adverse drug effects may be misconstrued as signs of CCD.[1,4,44]

TREATMENT

Diet

In both human AD and CCD of dogs, diet and dietary supplements have a substantial impact on both the development and progression of cognitive decline. Both dietary risk factors and preventive factors have been identified for AD in people, and these are suspected to be similar in CCD (**Table 1**).[1,3,4,27,45–48]

For human AD prevention and treatment, general dietary recommendations include the following: high intake of plant-based foods, antioxidants (and foods high in these substances), probiotics, soy beans, omega-3 polyunsaturated fatty acids (and foods rich in these substances), whole grains, fruits, vegetables, nuts, food enriched with medium-chain triglycerides (MCTs), and fish. Dietary choices that are likely to increase the risk of cognitive decline include red meats, poultry, refined sugar, processed food, and high-fat dairy products.[45,46] In addition to the general dietary recommendations

Table 1
Foods that have been identified to influence Alzheimer disease in people

Likely to Prevent or Treat AD in People	Likely to Increase Risk of AD in People
High intake of plant-based foods	Red meat or poultry
Antioxidants	Refined sugar
Probiotics	Processed food
Soy beans	High-fat dairy products
Omega-3-polyunsaturated fatty acids	
Whole grains	
Fruits/vegetables/nuts	
Medium-chain triglyceride–rich foods	
Fish	

and supplements already mentioned, other nutrients are recommended for cognitively impaired patients (**Box 6**).

These nutrients are often deficient in such individuals and include B vitamins (eg, B_{12}), vitamin C, vitamin E, mitochondrial cofactors (eg, L-carnitine, DL-α-lipoic acid), and carotenoids from green leafy vegetables. The addition of carotenoids and flavonoids as natural antioxidants from fruits, and particularly vegetables, has been associated with improvements in CCD.[4,45,46,49–51] These antioxidants are much more than just antioxidant in their actions. They can also act as mitochondrial cofactors and increase cellular endogenous antioxidant upregulation.[52] Therapeutic approaches using MCTs and natural antioxidants with essential vitamins and minerals seem to be effective. The 2 commercial CCD diets with the most evidence of efficacy are Hills b/d and Purina One Vibrant Maturity 7+ formula in clinical trials of CCD. Hills b/d was a foundational dietary approach, demonstrating that laboratory dogs with CCD showed better learning in a laboratory environment using tasks designed to test special awareness and object recognition primarily.[49–51] In addition, a clinical study using client-owned dogs showed some behavioral improvements compared with a control population on a placebo food. This diet was enriched with carnitine, lipoic acid, long-chain omega-3 polyunsaturated fatty acids, vegetable-based carotenoids, vitamin E, and vitamin C.[53]

MCTs are recommended in dogs with CCD, either as part of a formulated diet or a supplement, for several reasons. The AD/CCD brain has an impaired ability to use glucose, the brain's main energy source; MCTs provide an alternative energy source for the brain in cognitively impaired patients. These ketone bodies are rapidly converted to glucose precursors through interactions of astrocytes and glial cells with the surrounding neurons. MCTs may also improve brain mitochondrial activity and decrease the level of brain amyloid precursor protein.[4,54] MCTs are commonly included in ketogenic diets for both people and dogs. In one study, a proprietary mix of MCTs was shown to improve cognitive function in aged beagle dogs when incorporated into commercial food at 5% of the dry matter.[55]

Although the underlying premise regarding MCTs generating alternative energy sources for neurons through astrocyte metabolic alteration holds validity in humans, the premise is suspect in dogs. In humans there is considerable generation of β-hydroxybutyrate during metabolism of MCTs, whereas in dogs the generation of β-hydroxybutyrate is obscure, being between 20% and 70% of what is observed in people.[55–57] The lack of consistent ketone generation suggests that other mechanisms may play a role and it has been shown that the individual medium-chain fatty acids that increase in the bloodstream of rodents and people can affect neuronal function. The MCTs generally discussed are decanoic acid (10 carbons long, also known as capric acid) and octanoic acid (8 carbons long, also known as caprylic acid). Decanoic acid has been shown to dampen or block seizure activity in rodents because of direct inhibition of the AMPA (α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid) receptor, but more importantly may be involved in mitochondrial biogenesis

Box 6 **Supplements recommended for people affected by Alzheimer disease**
Vitamins B (B_{12}), C, and E
Mitochondrial cofactors (L-carnitine and DL-α-lipoic acid)
Carotenoids (green leafy vegetables)

through induction of PPARγ (peroxisome proliferator-activated receptor gamma), which may be important for neuronal survival.[58] Octanoic acid does not seem to have any action potential inhibitory activities from a neuronal perspective, but does seem to be easily metabolized by astrocytes, thereby allowing more local ketone production and promotion of the astrocyte lactate shuttle providing potential energy for neuronal metabolism.[58] Serum fatty acid derivatives of MCTs would be expected to increase from therapeutic trials; however, recent examination of food supplemented with 5% dry matter as a proprietary blend of MCTs that was proved to improve cognitive performance showed no increase in serum MCTs, whereas other less well-recognized fatty acid intermediates were enriched in the blood.[57] Foods rich in MCTs are limited, with only coconut and palm kernel oils being high enough to be used as sources of mainly lauric acid (12-carbon MCT) at nearly 50% of the triacylglycerols in these oils, with only 12% octanoic and decanoic acid in coconut oil and half this amount in palm kernel oil. Although coconut oil is used as the primary means of providing MCTs, it may not provide the appropriate MCTs; however, lauric acid, although not as well studied, does seem to have some antiinflammatory and antiamyloid properties as well.[59,60] To get the highest dose of MCTs, it may be best to purchase nutraceutical MCT oils, which are made from coconut and palm kernel oil but are pure MCTs at 100% of the oil. Proponents of using these oils suggest using them at approximately 5% of the dry matter in the diet initially by either providing as top dressing on a commercial food or providing primarily coconut oil as a fat source in a properly formulated home-prepared diet. This amount of coconut or MCT oil addition provides between 10% and 5% of the caloric content of the diet respectively. This additional oil can begin to create imbalance in the diet regimen, particularly if extra treats are being given as well. Therefore, addition of this oil should be conservative; **Table 2** shows relative quantities of MCT or coconut oil that can safely be provided, with consideration that these extra calories replace some of the calories consumed from the average commercial pet food. If home-prepared diets are warranted, it is best to have a board-certified veterinary nutritionist consulted to help formulate an appropriately balanced diet, particularly if a more ketogenic approach is desired.

Drugs and Nutraceuticals

In addition to dietary manipulation, there are numerous proposed therapeutic approaches to CCD, with variable evidence of efficacy in improving cognitive function and/or delaying progression of cognitive decline (**Box 7**).

Table 2
Calculated metabolizable energy for average elderly dog with potential amounts of coconut or medium-chain triglyceride oils that can be used to help with canine cognitive dysfunction

Dog Weight (kg)	ME (Kcal)[a]	10% ME Coconut Oil	5% ME MCT Oil	10% ME MCT Oil
5	335	1 tspn (35 kcal)	0.5 tspn (18 kcal)	1 tspn (35 kcal)
10	570	1.5 tspn (53 kcal)	0.75 tspn (27 kcal)	1.5 tspn (53 kcal)
20	950	3 tspn (105 kcal)	1.5 tspn (53 kcal)	3 tspn (105 kcal)
30	1280	3.5 tspn (123 kcal)	1.75 tspn (62 kcal)	3.5 tspn (123 kcal)
40	1590	4.5 tspn (158 kcal)	2.25 tspn (79 kcal)	4.5 tspn (158 kcal)
50	1880	5 tspn (175 kcal)	2.5 tspn (88 kcal)	5 tspn (175 kcal)

Abbreviations: ME, metabolizable energy; tspn, teaspoon.
[a] ME calculated from average ME for low-activity elderly dog – 100 $(kg)^{0.75}$.

Box 7
Nutraceuticals and drugs used for treatment of canine cognitive dysfunction

Omega-3-polyunsaturated fatty acids

MCTs

Phytochemicals (curcumin, resveratrol, green tea catechins)

S-adenosylmethionine

Phosphatidylserine

L-Deprenyl oral (selegiline, Anipryl, Atapryl, Carbex, Eldepryl, Zelapar)

Levetiracetam

Antiinflammatory drugs (eg, carprofen)

Apoequorin

The use of omega-3 fatty acids and MCTs as supplements is discussed earlier. Several cholinesterase-inhibiting drugs are used in patients with AD, but these have shown limited and transient efficacy; they are often also associated with gastrointestinal side effects.[61,62] Dosage and safety information for these drugs in dogs is lacking. The use of oral L-deprenyl (selegiline; Anipryl, Atapryl, Carbex, Eldepryl, Zelapar), an irreversible inhibitor of monoamine oxidase B, has been purported to improve cognitive function and slow progression of the disease in most dogs with CCD. However, there is considerable variability in the degree of response achieved among patients. L-Deprenyl is thought to exert its beneficial effects in the brain by restoring dopaminergic balance as well as enhancing catecholamine levels and decreasing levels of damaging free radical species. Most patients show a positive response within the first month of therapy. Despite apparent positive responses of patients with CCD to selegiline, there is evidence that this drug does not have a significant effect on cognitive function in these patients or in people with AD. The clinical efficacy studies supporting selegiline use in CCD are based primarily on owner response to questionnaires rather than on standardized comparative cognitive testing procedures of treated and untreated patients. Because selegiline may produce nonspecific low-level hyperactivity by increasing brain catecholamine levels, the response observed by owners may not truly be representative of improved cognitive ability. Selegiline is not considered an effective drug for human AD because of variable responses and overall minimum improvement of cognitive function.[1,4,62–64] Recent evidence suggests that the anticonvulsant drug levetiracetam is effective in improving cognitive function and decreasing hyperexcitability in AD and animal models of AD; potential mechanisms of action include improved central nervous system mitochondrial function (and associated improved synaptic transmission) and inhibition of oligomeric Aβ-induced astrocyte glutamate release.[65–67] Behavioral changes (particularly anxiety) in patients with CCD may be alleviated with the use of GABAergic drugs, such as gabapentin or pregabalin. Because inflammatory changes have been identified in the brains of dogs with CCD, the use of antiinflammatory drugs (eg, carprofen) has also been proposed. Some naturally occurring phytochemicals (eg, curcumin, resveratrol, green tea catechins) may hold some promise as treatment options for CCD.[1,4] Oral S-adenosylmethionine (SAMe) was shown to be effective in improving clinical signs of mental decline in dogs with CCD in 1 randomized, placebo-controlled study.[68] Phosphatidylserine, a membrane phospholipid, showed some efficacy in improving cognitive function in patients with CCD in several clinical trials.[1,4,69,70] Apoequorin is a calcium-buffering protein

Box 8 Nutraceuticals and drugs used to suppress the clinical signs of canine cognitive dysfunction (anxiety and abnormal sleep-wake cycles)
Melatonin
Valerian root
Dog-appeasing pheromone

derived from jellyfish that has been shown experimentally to protect against ischemic neuronal cell death. Apoequorin has shown efficacy in improving cognition in older dogs and people, including people with mild cognitive impairment. A large number of complementary therapies have been suggested for the treatment of CCD, with the primary goals of calming the patient, reducing anxiety, and normalizing the sleep-wake cycle (**Box 8**). These therapies include melatonin, valerian root, dog-appeasing pheromone, docosahexaenoic acid(an omega 3 fatty acid), and various antioxidants and mitochondrial cofactors.[1,4,44]

Cognitive Enrichment

Cognitive enrichment, such as regular exercise, social interactions, and introduction of new toys, has been shown to improve cognitive function in dogs with CCD and prevent or delay cognitive decline in dogs as they age.[1,4,44,71] Similarly, cognitive enrichment has been proved to protect against cognitive decline and improve cognitive function in older people and human patients with AD, respectively. People who are more intellectually stimulated earlier in life (eg, education level, complexity level of work and/or leisure activity) and who exercise regularly are less likely to experience cognitive impairment in later years. This phenomenon has been shown even in older people with evidence of brain disorder. Instituting cognitive enrichment strategies such as regular physical exercise has been shown to improve cognitive function in people with AD.[71–73] Similar evidence has been accumulated in rodent AD experimental models.[71] Several mechanisms are thought to be involved in the beneficial effects of cognitive enrichment. These mechanisms include improved neuronal plasticity (with formation of new synapses) and decreased hippocampal neuronal loss. Related to these mechanisms is a concept of cognitive reserve. Cognitive reserve refers to a hypothesis that the brain can use available neural pathway networks as backup systems in case of brain disorder, as occurs with CCD/AD. This cognitive reserve is likely well developed in individuals who have experienced cognitive enrichment throughout early life but can also be developed to some extent in older patients showing signs of cognitive decline.[71,73]

Dog owners can provide cognitive enrichment to their pets in several ways. In addition to home-based methods (eg, regular walks, regular introduction to new toys), there are an increasing number of pet rehabilitation facilities available that can provide such enrichment on a recurring basis (**Fig. 4**).

Chinese Herbal Therapy

In recent years, Chinese herbal therapy has been investigated extensively in experimental (rodent) AD models, as well as in human AD clinical trials. There is evidence of efficacy for many single herbs, molecular compounds from such herbs, as well as herbal formulas. The mechanisms of action in improving cognitive function for many of the individual herbs or their chemical constituents have been elucidated.

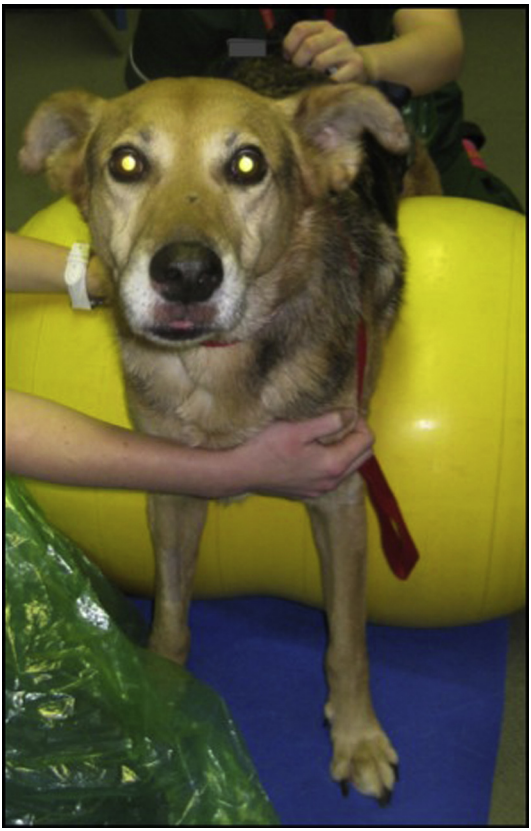

Fig. 4. Dog in rehabilitation using an inflatable peanut ball.

These mechanisms include dissolution of Aβ in the brain, inhibited generation of brain Aβ inclusions, reduction of brain acetylcholinesterase activity, antioxidant activity, protection of brain mitochondria from oxidative stress, protection against glutamate-mediated neurotoxicity, antiinflammatory activity, improvement in brain blood flow, improvement in neuronal glucose use, improvement in neuronal ATP production, and inhibition of neuronal apoptosis in the brain.[74–92] The list of individual Chinese herbs with evidence of efficacy for treating cognitive impairment is extensive. The most prominent such herbs in this category, in addition to their relevant molecular mechanisms of action, are summarized in **Table 3**.

In addition to single Chinese herbs, several herbal formulas have shown efficacy in the treatment of cognitive impairment in both experimental rodent models and people with AD. There are several studies showing efficacy of Liu Wei Di Huang (also known as Rehmannia 6) in the treatment of cognitive impairment.[88,90–92] Human clinical studies of Sai Luo Tong formula, which consists of the bioactive components of the 3 herbal extracts from *Panax ginseng*, *Ginkgo biloba*, and *Crocus sativa*, showed significant improvement in neurocognitive function, learning, and memory.[93,94] Other Chinese herbal formulas such as Fu Fang Dan Shen, Ba Wei Di Huang Wan, and Yi Gan San, have also shown encouraging data on treating cognitive and psychological symptoms in patients with dementia.[95–97]

Table 3
Individual Chinese herbs and their relevant molecular mechanisms

Latin and Chinese (in Parentheses) Herbal Name	Mechanism of Action
Ginkgo biloba (Bai Guo)	Antioxidant; antiapoptotic; antiinflammatory
Huperzia serrata (Qian Ceng Ta)	Procholinergic; antiglutamate (NMDA antagonism)
Curcuma longa (Yu Jin)	Inhibits Aβ production; antioxidant; antiinflammatory
Ginseng (Ren Shen)	Antioxidant; antiglutamate (NMDA antagonism); decrease Aβ
Coptis chinensis (Huang Lian)	Inhibits Aβ production; procholinergic; antioxidant; antiinflammatory
Polygala tenuifolia (Yuan Zhi)	Inhibits Aβ production; antiapoptotic
Salvia miltiorrhiza (Dan Shen)	Inhibits Aβ aggregation; antioxidant; antiinflammatory
Angelica sinsensis (Dang Gui)	Antioxidant; antiinflammatory
Crocus sativus (Zang Hong Hua)	Antioxidant; antiapoptotic
Gastrodia elata (Tian Ma)	Procholinergic; antioxidant
Rehmannia glutinosa (Shu Di Huang)	Procholinergic; antioxidant
Epimedium (Yin Yang Huo)	Antioxidant; antiapoptotic
Magnolia officinalis (Xin Yi Hua)	Procholinergic; antioxidant
Scutellaria baicalensis (Huang Qin)	Inhibits Aβ aggregation; antioxidant; antiinflammatory; antiapoptotic
Camellia sinensis (Cha Hua)	Reduces Aβ levels; antioxidant; antiinflammatory; antiapoptotic

Abbreviation: NMDA, *N*-methyl-D-aspartate.

Acupuncture

Although not yet clinically investigated as a therapeutic modality in dogs with CCD, there is considerable evidence in rodent AD models and human AD clinical trials supporting a beneficial role for acupuncture (particularly electroacupuncture) in the treatment of cognitive dysfunction.[98–101] Several likely mechanisms of action for the positive effects of acupuncture have been elucidated from experimental rodent AD models (**Box 9**).

These mechanisms include enhanced neuronal glucose use (as verified by PET scans), decreased accumulation of Aβ in the brain (via inhibition of the mTOR [mammalian target of rapamycin] pathway), increased production of neurotrophic factors (such as brain-derived neurotrophic factor [BDNF]) with subsequent proliferation of neuronal stem cells, and protection from or reversal of synaptic loss and dendritic atrophy in the hippocampus. Overall, the various mechanisms that have been uncovered experimentally are thought to enhance brain neuroplasticity and improve cognitive function.[98–100,102–108]

Electroacupuncture stimulation at Yin-tang and LI-20 significantly ameliorated the learning and memory deficits of AD mice (detected with maze tests) and was accompanied by increased BDNF and decreased amyloid formation of Aβ proteins (biomarker of CDS) in the brain. Acupuncture at GV-20, GV-14, and ST-36 increased expression of synaptophysin and neurotrophin in the hippocampal region; acupuncture at HT-7 and PC-6 promoted cholinergic neural transmission; acupuncture at

Box 9
Mechanism of action for positive effects seen with acupuncture

Enhanced neuronal glucose use

Decreased accumulation of Aβ in the brain

Increased production of neurotrophic factors

Proliferation of neuronal stem cells

Protection or recovery from synaptic loss and dendritic atrophy in the hippocampus

GV-14, GV-20, CV-6, CV-12, CV-17, SP-10, ST-36, HT-9, KID-1, LU-11, PC-9, GB-34, and LIV-3 suppressed oxidative stress and attenuated apoptosis; acupuncture at GB-34, GV-20, LIV-3, SP-10, and ST-36 reduced the expression of microglia; and acupuncture at LI-4 and LIV-3 activated regions of the cerebrum responsible for memory and cognition.[109] A human clinical trial confirmed that acupuncture can enhance hippocampal activity in patients with AD. The study used 14 patients with AD and 14 healthy elders. Stimulation of acupoints LI-4 and LIV-3 was evaluated with functional MRI during baseline, resting, and after stimulation. In the resting state, several frontal and temporal regions (cognitive-related regions) showed decreased hippocampal connectivity in patients with AD relative to control subjects. Following acupuncture, patients with AD showed increased connectivity in most of the hippocampal-related regions.[110]

SUMMARY

CCD is a common neurodegenerative condition of aging dogs with many similarities to AD of people. The pathophysiology of CCD/AD is multifaceted, involving many interrelated biochemical aberrations. Central to the development of CCD/AD are compromised brain vasculature and the deposition in the brain parenchyma of a toxic protein called Aβ. These two central processes are intertwined, because each can promote progression of the other. Although dogs with CCD are typically recognized as showing cognitive impairment by 8 years or older, there is considerable evidence that the processes leading to this clinical state begin earlier in life. Pet owners and veterinarians should therefore consider preventive measures against the development of CCD when dogs are middle aged. Diagnosis of CCD often depends on an accurate history of the patient's clinical progression, derived from the dog owner. It is important for veterinarians to maintain a high index of suspicion for CCD and ask the right questions. It is important for dog owners and veterinarians to realize that cognitive impairment or senility should not be considered a normal part of the canine aging process. Because CCD-suspect dogs tend to be elderly, other disease processes that could mimic CCD need to be ruled out. Although not commonly pursued, brain MRI of dogs with CCD usually shows a small interthalamic adhesion thickness. Dogs with CCD usually do not reach the same level of disease severity as their human AD counterparts. These patients also usually respond well to therapeutic interventions and often have a good long-term quality of life. Several therapeutic approaches have been shown to be effective in treating CCD, principal among them being dietary modulations, a variety of nutraceutical supplements, and cognitive enrichment. The anticonvulsant drug levetiracetam has been shown to be effective in experimental models of AD as well as human AD clinical trials in improving cognitive function. This therapy should be considered for treatment of CCD as well. Recent literature in rodent AD models and

human clinical AD trials suggests an important role for Chinese herbal therapy and acupuncture in the treatment and prevention of cognitive dysfunction.

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